Lecture 11: Gas & Ice Giants
Jupiter, Saturn, Uranus, Neptune

Planetary Systems

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Jupiter’s south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS

Learning Objectives

By the end of this lecture, you will be able to:

- Compare the internal structures of the four giant planets.
- Describe atmospheric dynamics on Jupiter, Saturn, Uranus, and Neptune.
- Outline the diversity of giant-planet satellites and ring systems.
- Derive the Roche limit and apply it to Saturn's rings.
- Use the gas-giant / ice-giant dichotomy as a laboratory for exoplanet analogues.

Four giant planets, two physical classes, hundreds of moons, and only one mission ever to either ice giant.

Recap: Where We Are in the Course

- L9 (Earth & Venus): rocky-planet near-twins with divergent surface evolutions.
- L10 (Mercury & Mars): rocky-planet limiting cases (smallest/densest vs. low-mass volatile-rich).
- Today: the four giant planets, >99.5% of all planetary mass beyond the Sun.
- L12: small bodies; L13: exoplanets; L14: synthesis.

1. Gas Giants and Ice Giants: An Overview

Solar system family portrait, Voyager 1. Credit: NASA/JPL-Caltech

Two Classes, Four Planets

- **Gas giants:** Jupiter, Saturn
  - H/He envelopes dominate mass
  - 318 and 95 $M_\oplus$
  - Together: ~92% of all planetary mass
- **Ice giants:** Uranus, Neptune
  - H/He only ~10–20% of mass
  - Bulk = water, ammonia, methane fluids
  - 14.5 and 17.1 $M_\oplus$

- No solid surface; ”1 bar level” is convention.
- Smooth gas → supercritical fluid → metallic plasma.
- Rapid rotation: ~ 10 h, oblate by 6–10%.

Energy Budgets: Internal Heat Excess

- Jupiter: emits ~1.7× absorbed solar flux → Kelvin-Helmholtz contraction.
- Saturn: ~1.8× → helium-rain release of gravitational energy.
- Uranus: ~1.06× → *anomalously low*; the major outlier.
- Neptune: ~2.6× → source still uncertain.

Three of four giants are net energy exporters; Uranus is the puzzle that drives mission concepts.

Why Rapid Rotation Matters

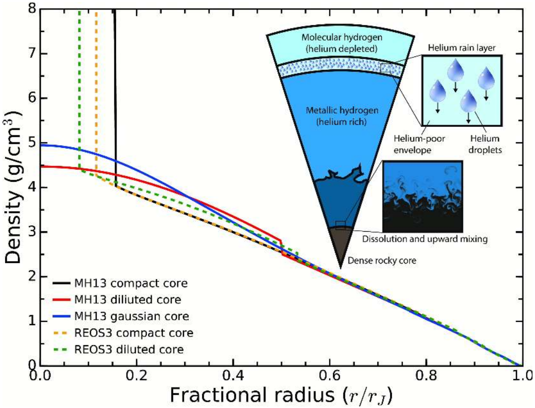
- Jupiter: 9 h 56 min; Saturn: 10 h 33 min; rocky planets at 24 h or longer.
- Strong Coriolis → latitude-aligned bands of zonal wind.
- Oblateness encodes interior density via gravity harmonics $J_2, J_4, J_6, \ldots$.
- Juno (Jupiter) and Cassini Grand Finale (Saturn) measured J_n to high order.
- Equatorial jet speeds counterintuitively *increase* with distance from Sun.

2. Jupiter: Interior and Atmosphere

Jupiter’s south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS

Jupiter Interior: Layered Fluid

- Outer molecular H_2 + He envelope
- Transition to metallic hydrogen at ~100 GPa, ~5000 K
- $r/R_J \approx 0.85$ for the conducting transition
- Dynamo source: metallic hydrogen
- Centre: ~4000 GPa, ~20 000 K

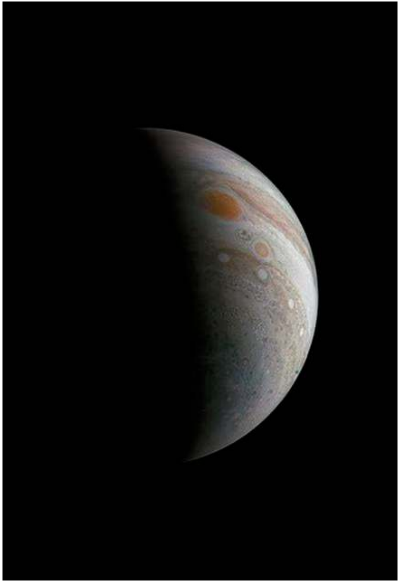


Wahl et al. (2017)

The Dilute Core: A Decade-Old Revolution

- Pre-Juno: assumed compact, well-defined ~10 $M_\oplus$ core.
- Juno gravity (J_4, J_6, J_8, J_{10}) cannot match compact-core models.
- Juno data fit naturally with heavy elements distributed across the inner 30–50% of the planet.
- ”Dilute” / ”fuzzy” core picture: Wahl+ 2017, Militzer+ 2022.
- Total heavy-element content: $\sim$ 20–40 $M_\oplus$.

Dilute core implies efficient mixing or non-standard accretion; rewrites textbook formation picture.

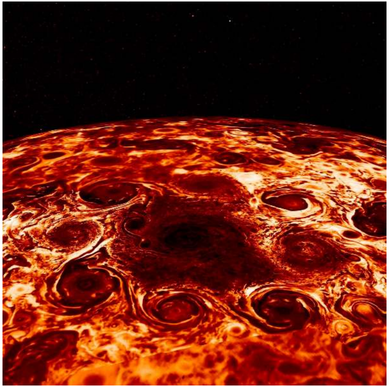


Crescent Jupiter and Great Red Spot, JunoCam (perijove 3, Dec 2016). Credit: NASA/JPL-Caltech/SwRI/MSSS / R. Tkachenko

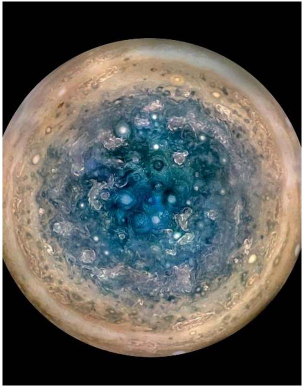
Reading the Jupiter Atmosphere Image

- Cloud decks at 0.1–10 bar: $\mathrm{NH_3}$ (top), $\mathrm{NH_4SH}$, $\mathrm{H_2O}$ (deepest).
- ~15 alternating zonal jets pole-to-pole; peak ~180 m s^{-1}.
- Bright zones = rising air, ammonia clouds; dark belts = descending haze.
- Great Red Spot: anticyclonic vortex observed since at least 1830.
- GRS shrinking: ~40 000 km in 1900 → ~14 000 km today.
- Galileo probe (1995): confirmed layering but found a dry entry site.

Polar Cyclones



North pole: 1 + 8 cyclones, Adriani+ 2018



South pole: 1 + 5 cyclones

Stable polygonal arrangements persist across many Juno orbits. Polar weather not seen at high resolution before 2016.

Jets, Aurorae, and the Great Blue Spot

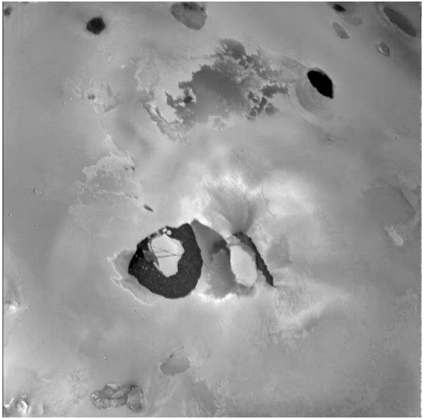
- Kaspi+ 2018: equatorial jets penetrate to ~3000 km depth (gravity asymmetries).
- Below: magnetic stress damps differential rotation; deep interior → solid-body.
- Aurorae are the strongest in the solar system.
- Driven by magnetosphere + Io mass loading (~1 t s^{-1} of S, O).
- Auroral footprints of Io, Europa, Ganymede directly imaged in UV.
- "Great Blue Spot": isolated equatorial magnetic anomaly mapped by Juno (Connerney+ 2022).

3. The Galilean Moons

Europa, Galileo orbiter mosaic. Credit: NASA/JPL-Caltech/SETI Institute

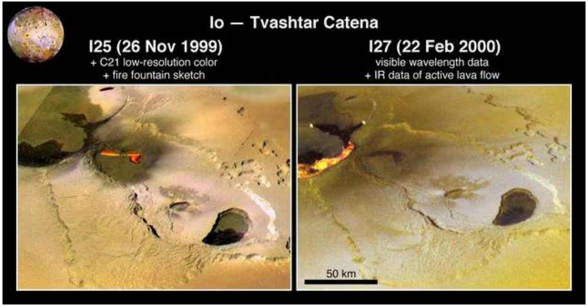
Io: Most Volcanic Body in the Solar System

- ~400 active volcanic centres
- Heat output ~ 10^{14} W (tidal, not radiogenic)
- Driven by 1:2:4 Laplace resonance with Europa, Ganymede
- Predicted by Peale+ 1979, days before Voyager 1
- Surface resurfaced on ~Myr timescales



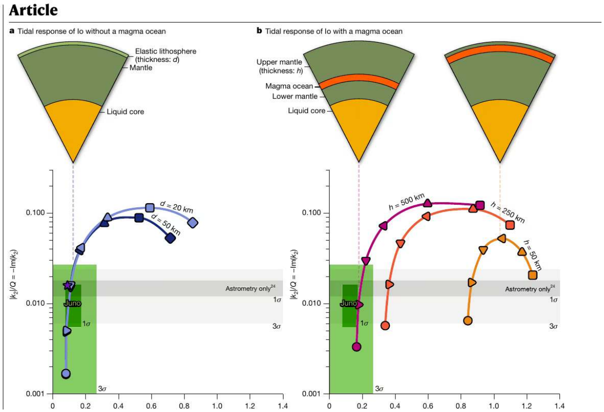
Loki Patera lava lake, Voyager 1 (1979). NASA/JPL

Io: Tvashtar Eruption

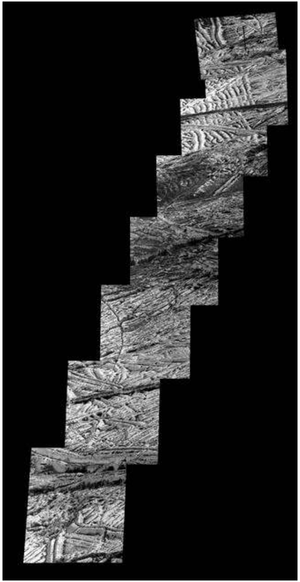


- Tvashtar Catena fire fountain imaged by Galileo (orbits I25, Nov 1999 and I27, Feb 2000).
- Composite of low-resolution colour, visible imaging, and IR detection of active lava flow.
- Tenuous $\mathrm{SO_2}$ atmosphere: sublimes by day, freezes at night.
- Strong day-night asymmetries.

Io Interior: Juno Says No Magma Ocean



- Juno close flybys 2023–2024.
- Two-way Doppler → tidal Love number $k_2 \approx 0.125 \pm 0.047$.
- **Too small** for global shallow magma ocean.
- Mostly solid silicate mantle with localised partial melts.
- Park et al. (2024).

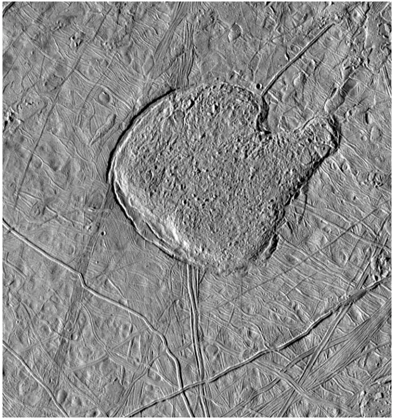


Europa surface, Galileo orbiter. Credit: NASA/JPL-Caltech/SETI Institute

Reading Europa: Lineae and Chaos

- Surface dominated by water ice; criss-crossed by long fractures (lineae).
- Patches of ”chaos terrain”: ice has foundered, broken up, refrozen.
- Crater counts: surface age only 40–90 Myr (Pappalardo+ 1999).
- Diameter 3122 km (smaller than Moon).
- Strong tidal heating sustained by Laplace resonance with Io and Ganymede.

The Case for Europa's Ocean



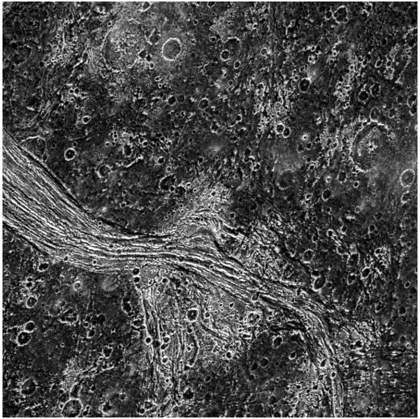
Conamara Chaos, Galileo PIA01640

- Chaos morphology → episodic basal melting.
- Galileo magnetometer: induced field requires global conductor (Khurana+ 1998).
- Hubble UV: candidate plumes near south pole (Roth+ 2014, Sparks+ 2017).
- Modern picture: 15–25 km ice over ~100 km ocean, ~Earth-ocean volume.

Ganymede: Largest Moon, Only Moon-Dynamo

- Radius 2634 km: bigger than Mercury.
- Only moon known to host an intrinsic dynamo (Galileo, 1996).
- Fully differentiated: Fe core → silicate mantle → icy shell.
- Ocean: ~100 km deep below ~150 km ice (Saur+ 2015 from HST aurorae).

Ganymede: Grooved Terrain

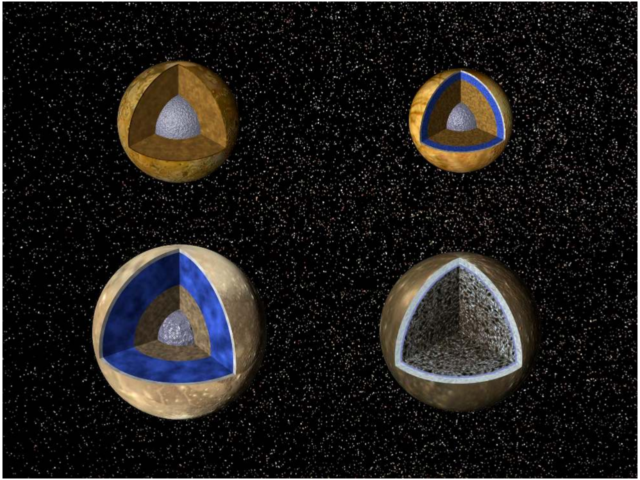


- Lagash Sulcus: bright grooved terrain cuts dark cratered terrain.
- Records ancient tectonic extension.
- Distinct from Callisto's intact, undisturbed surface.
- ESA's JUICE will orbit Ganymede in 2034.

Callisto: The Quiet Sibling

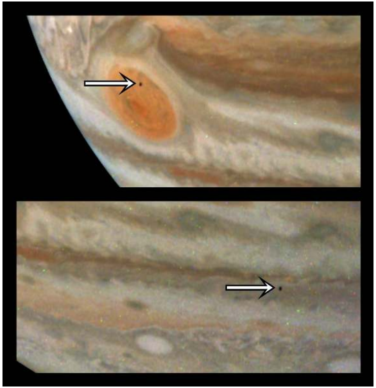
- R = 2410 km, density 1834 kg m^{-3}.
- $C/MR^2 \approx 0.355$: only *partially* differentiated (Anderson+ 2001).
- No tectonic resurfacing; outside the 1:2:4 Laplace lock.
- Yet: induced field signal → probable subsurface ocean.
- Lower radiation than inner Galileans ("boring but habitable").

Galilean Comparative Anatomy



Interior structures (l-r: Io, Europa, Ganymede, Callisto). NASA/JPL artist concept (PIA01082).

Jupiter's Rings and Small Moons



Amalthea silhouette against Jupiter, Juno PJ59 (2024).

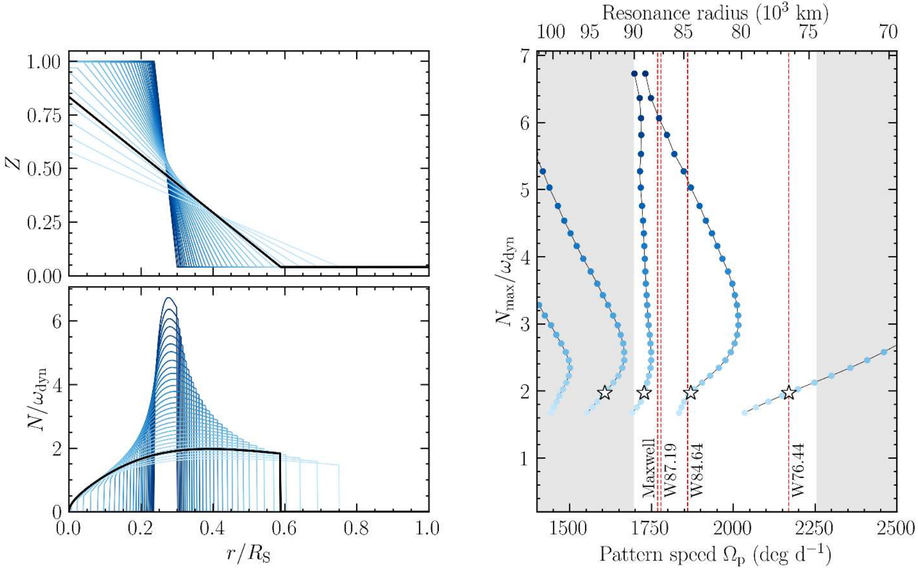
- Faint dusty rings, discovered by Voyager 1 (1979).
- Sourced by impact gardening of Amalthea, Adrastea, Metis, Thebe.
- Continuously replenished, transient → *not* a long-lived disk.

4. Saturn: Interior and Rotation

Saturn’s main rings, Cassini. Credit: NASA/JPL-Caltech/SSI

Saturn Interior: Lower Pressures, Helium Rain

- Mass 95 $M_\oplus$; central pressure ~ 1/3 Jupiter's.
- Molecular-metallic hydrogen transition deeper, milder conditions.
- Cassini Grand Finale (2017): gravity field rivalling Juno's at Jupiter.
- Helium becomes immiscible in metallic H at 1–3 Mbar, 5–10 kK.
- Droplets fall, release gravitational PE → closes Saturn's energy budget.
- Predicts He depletion in upper envelope; consistent with Voyager.



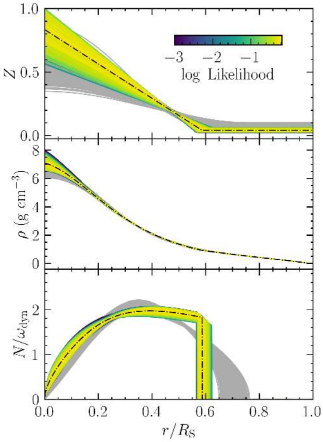
Kronoseismology constraints on Saturn's interior, Mankovich & Fuller (2021).

Reading the Kronoseismology Result

- Saturn's f-mode oscillations excite density waves in the C ring at resonance radii.
- Pattern speeds matched to wave positions in 75 000–95 000 km range.
- Match requires *stably stratified*, *dilute* heavy elements out to ~ 60% of R_S.
- Total heavy-element mass ~ 17 $M_\oplus$.
- First seismic constraint on *any* giant planet interior.

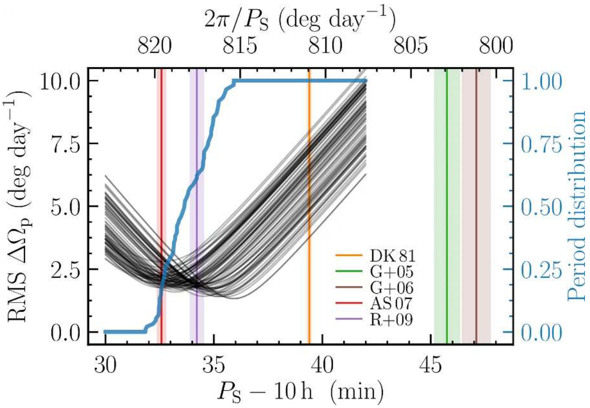
Like Jupiter, Saturn has a fuzzy core, not a compact one. Both giants share the same surprise.

Heavy-Element Distribution Profile



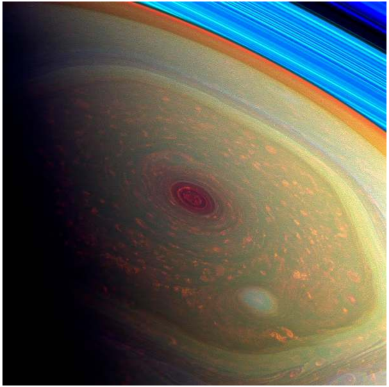
- Top: $Z(r)$ (heavy-element mass fraction).
- Middle: density $\rho(r)$.
- Bottom: Brunt-Väisälä frequency N.
- Yellow: maximum-likelihood profile.
- Mankovich & Fuller (2021).

Saturn's Rotation Period from Ring Seismology



- Saturn's magnetic dipole nearly axisymmetric → no clean radio period.
- C-ring f-mode wave residuals as a function of assumed *P*.
- Best estimate: P_S = 10 h 33 min 38 s.
- Mankovich+ 2019.

Saturn's Atmosphere and the Hexagon



- Same $\mathrm{NH_3/NH_4SH/H_2O}$ cloud decks as Jupiter, deeper and stretched.
- Banded contrast muted in visible.
- Hexagonal jet at ~ 78° N: standing Rossby wave, six sides.
- Continuous since Voyager (1981).
- Equatorial jet ~400 m s^{-1} (Cassini).

5. Saturn’s Rings

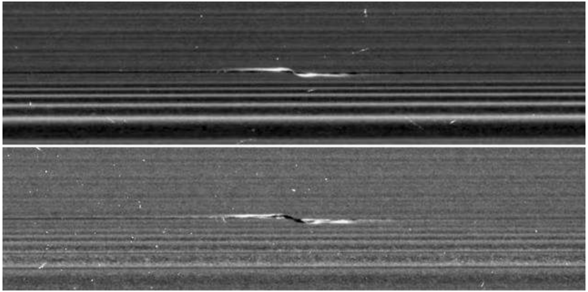
Saturn’s main rings (radial scan), Cassini. PIA08389

Natural-colour radial scan: C, B ring, Cassini Division, A ring. Cassini (PIA08389). NASA/JPL-Caltech/SSI

Reading the Ring Scan

- Main ring system spans 67 000–140 000 km from Saturn's centre.
- D, C, B rings inside the fluid Roche limit (computed below: ~126 000 km).
- Cassini Division at ~118 000 km: cleared by 2:1 resonance with Mimas.
- Outer A ring + F ring: just outside Roche; A-ring edge sharpened by 7:6 with Janus.
- Composition: >95% water ice, particle sizes 1 cm to 10 m.
- Vertical extent only ~10 m: thinnest known objects in the universe (relative to extent).

Embedded Moons and Propellers



- F ring shepherded by Prometheus + Pandora.
- Encke Gap held open by embedded moon Pan.
- "Propeller" features: ~1 km moonlets too small to clear gaps.
- Direct measurement of orbital migration in a disk (Tiscareno+ 2013).

The Rings Are Young

- Total ring mass ~ 1.5×10^{19} kg = 40% of Mimas (Iess+ 2019, Cassini gravity).
- Far too small for primordial origin: would have darkened by infall over 4.5 Gyr.
- Brightness + mass + composition → age ~100 Myr (Crida+ 2019).
- ”Ring rain”: water + organics infall onto Saturn’s atmosphere (Waite+ 2018).
- Ground-based H_3^+ ionosphere modulation confirms (O’Donoghue+ 2019).

Remaining lifetime ~ 10^8 yr; Saturn’s rings are a transient phase, not a permanent feature.

What Made the Rings?

- Wisdom+ (2022): tidal disruption of an icy Mimas-class moon scattered inside the Roche limit.
- Could simultaneously explain ring origin and the unusual mid-sized moon orbits.
- Alternative: late capture of a Kuiper Belt object.
- Charnoz+ (2009): possibly older but with continuous mass shedding.
- Pedagogical message: solar-system bodies do not all date from 4.5 Gyr ago.

6. Blackboard Derivation: The Roche Limit

Solar system family portrait. Credit: NASA/JPL-Caltech

Setup: Why Rings Exist

- Roche limit: distance inside which a fluid satellite's self-gravity loses to the planet's tidal stretch.
- Concept introduced by Roche (1849).
- Sets where rings can exist and where moons cannot accrete.

Configuration:

- Planet: mass M_p, radius R_p, density ρ_p.
- Satellite: mass M_s, radius R_s, density ρ_s, at distance d from planet centre.
- Assume $d \gg R_s$ so a Taylor expansion across the satellite is valid.
- Work in the satellite's rest frame.

Step 1: Tidal Acceleration Across the Satellite

Acceleration at the satellite's centre:

$$g_{\text{centre}} = \frac{GM_p}{d^2}$$

At the near point (closest to planet, distance $d - R_s$):

$$g_{\text{near}} = \frac{GM_p}{(d - R_s)^2}$$

Differential (tidal) acceleration that stretches the satellite:

$$\Delta a_{\text{tidal}} = g_{\text{near}} - g_{\text{centre}} = GM_p \left[\frac{1}{(d - R_s)^2} - \frac{1}{d^2} \right]$$

Step 2: Taylor Expand in R_s/d

For $R_s \ll d$:

$$\frac{1}{(d - R_s)^2} = \frac{1}{d^2}\left(1 - \frac{R_s}{d}\right)^{-2} \approx \frac{1}{d^2}\left(1 + \frac{2R_s}{d}\right)$$

So:

$$\boxed{\Delta a_{\text{tidal}} \approx \frac{2GM_pR_s}{d^3}}$$

Tidal forces fall off as d^{-3}, not d^{-2}. This is the universal scaling.

Step 3: Self-Gravity at the Satellite's Surface

Uniform density ρ_s:

$$a_{\text{self}} = \frac{GM_s}{R_s^2} = \frac{G \cdot \frac{4}{3}\pi R_s^3 \rho_s}{R_s^2} = \frac{4}{3}\pi G \rho_s R_s$$

Setting tidal = self-gravity at the disruption threshold:

$$\frac{2GM_p R_s}{d^3} = \frac{4}{3}\pi G \rho_s R_s$$

Step 4: Solve for d_R

Cancel G and R_s, substitute $M_p = \frac{4}{3}\pi R_p^3 \rho_p$:

$$\frac{2 \cdot \frac{4}{3}\pi R_p^3 \rho_p}{d^3} = \frac{4}{3}\pi \rho_s$$

Rearrange:

$$d^3 = 2R_p^3 \frac{\rho_p}{\rho_s}$$

$$\boxed{d_R^{\mathrm{rigid}} = 2^{1/3} R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3} \approx 1.26\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}}$$

Fluid vs. Rigid Roche Limit

- Rigid case: spherical, holds shape until disruption.
- Fluid case: deforms into ellipsoid first, disrupts further out.
- Roche's full fluid calculation gives prefactor ≈ 2.46:

$$\boxed{d_R \approx 2.46\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}}$$

Real bodies with finite material strength sit between the two limits.

Step 5: Apply to Saturn's Rings

$$R_p = 58{,}232\ \mathrm{km}, \quad \rho_p = 687\ \mathrm{kg\,m^{-3}}, \quad \rho_s \approx 1000\ \mathrm{kg\,m^{-3}}$$

$$d_R \approx 2.46 \times 58{,}232 \times (0.687)^{1/3}\ \mathrm{km} \approx 2.46 \times 58{,}232 \times 0.883$$

$$\boxed{d_R \approx 126{,}000\ \mathrm{km}}$$

Within ~10% of A-ring outer edge (~137 000 km). Discrepancy: ring-particle porosity, ice rigidity, density-wave structure.

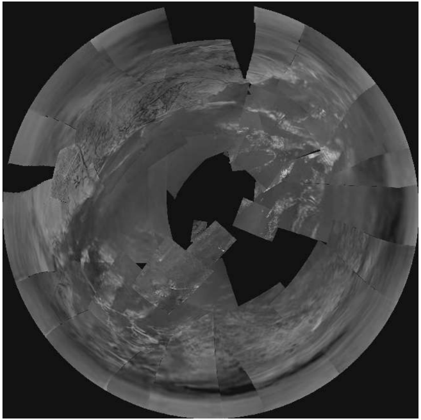
Why Roche Matters

- Inside d_R: differential pull > self-gravity → no agglomerate > 10 m holds together.
- Particles orbit as a collisional disk; cannot accrete into a moon.
- Outside d_R: same population can re-accrete (e.g., small inner moons of Saturn).
- Used: Saturn ring origin, Triton's eventual fate, comet Shoemaker-Levy 9 disruption (Jupiter, 1992).

7. Saturn’s Moons: Titan, Enceladus, and More

Titan’s polar lakes, Cassini RADAR. NASA/JPL-Caltech/ASI/Cornell

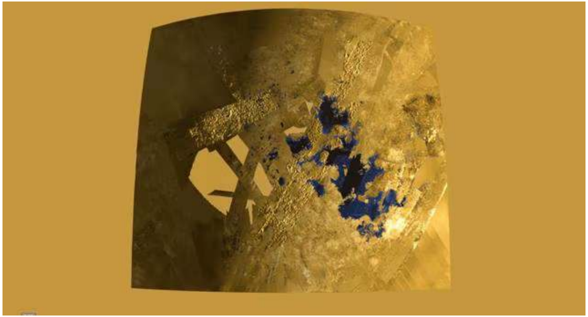
Titan: Atmosphere and Methane Cycle



Huygens descent imagery, 14 Jan 2005 (PIA07870)

- R = 2575 km; second largest moon.
- N_2 atmosphere: 1.5 bar surface, 4× Earth column density.
- Surface 94 K → methane near triple point.
- Methane: rain, rivers, lakes, evaporation cycle.

Titan: Polar Lakes

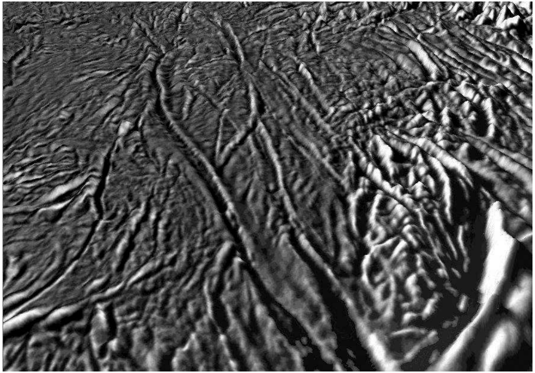


- Hundreds of lakes near the north pole (Cassini RADAR).
- Kraken Mare, Ligeia Mare ~Caspian-sized.
- Photochemistry produces "tholins": HCN, C_2H_2, ..., organic haze.
- Tholins settle → form equatorial dunes.

Titan: Subsurface Ocean

- Cassini gravity + tidal Love number → global subsurface water-ammonia ocean (Iess+ 2012).
- Ice shell ~100 km thick.
- Joins Europa, Ganymede, Callisto, Enceladus on the ocean-worlds list.
- NASA's **Dragonfly** (launch 2028, arrival 2034): first rotorcraft on another world.
- Targets Selk impact crater for organics + transient melt-water chemistry.

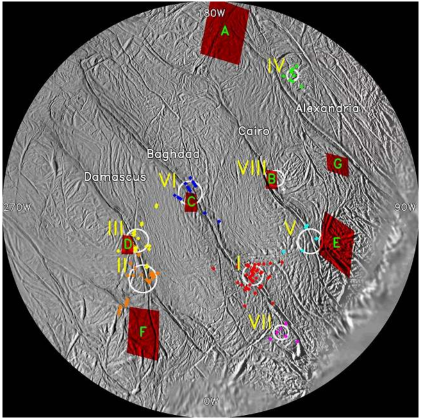
Enceladus: A Moon That Erupts



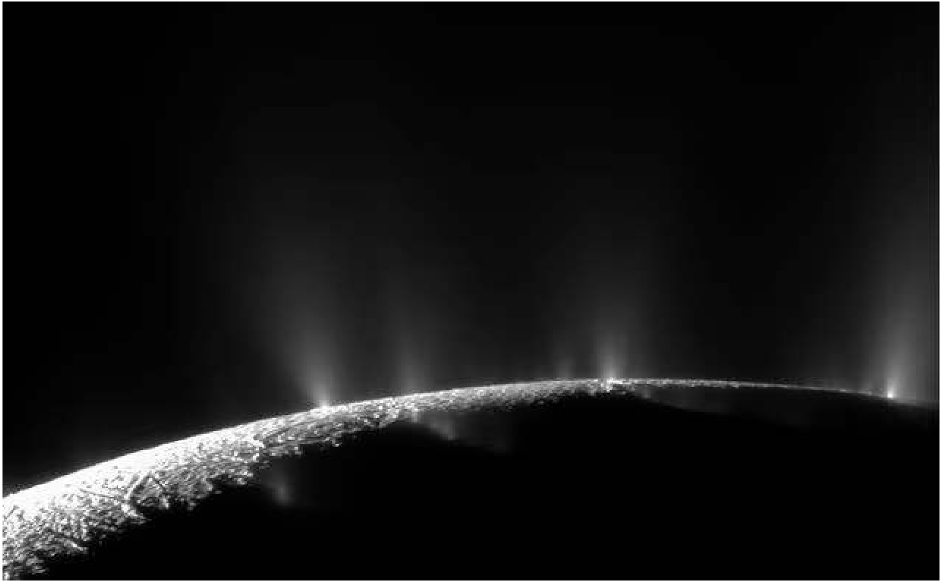
Tiger stripes (south polar fractures)

- R = 252 km, but oversized scientific impact.
- 2005 Cassini discovery: active geyser plumes from south-pole ”tiger stripes”.
- Porco+ 2006.
- Plume composition: 90% $\mathrm{H_2O}$, also $\mathrm{CO_2}$, $\mathrm{NH_3}$, salts, organics.

Enceladus: Heat at the Stripes



- Cassini CIRS: stripes 10s of K warmer than surrounding terrain.
- Jet sources align with hot spots.
- Total south-polar heat > 10 GW, far above radiogenic budget.
- Sustained by 2:1 resonance with Dione.



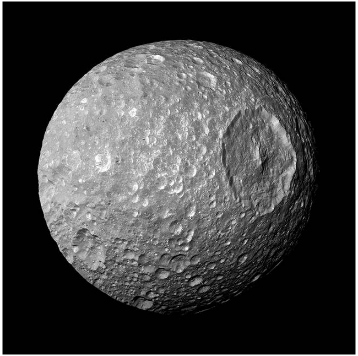
Dozens of geyser jets erupting from Enceladus’s tiger stripes. Cassini, NASA/JPL-Caltech/SSI.

Enceladus: Habitability Ingredients

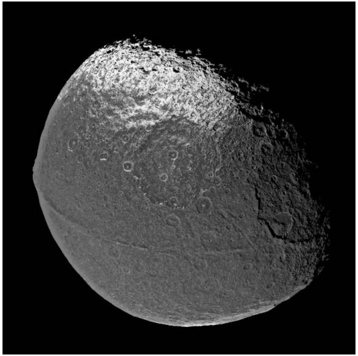
- Global ocean below ~20–30 km ice shell.
- Plume H_2 (Waite+ 2017) → thermodynamic disequilibrium, hydrothermal-vent analogue.
- Silica nanoparticles (Hsu+ 2015) → rock-water reactions today.
- Phosphates (Postberg+ 2023) → key biochemistry building block in seawater.
- Cassini's mass spectrometers were not designed to detect biosignatures.
- 2022 Decadal Survey priority: future Enceladus orbilander or sample return.

Most accessible ocean-world habitability target; plumes deliver the ocean to your spacecraft.

Mid-Sized Moons: Mimas and Iapetus

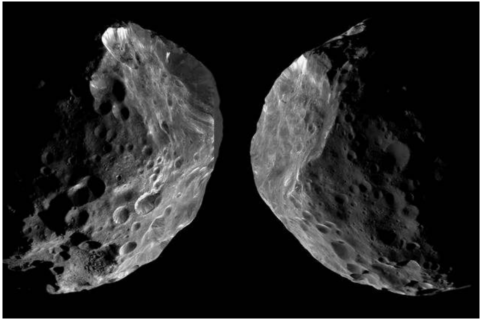


Mimas, Herschel crater (close to disruption threshold)



Iapetus, two-toned hemispheres

Phoebe: A Captured Kuiper Belt Object



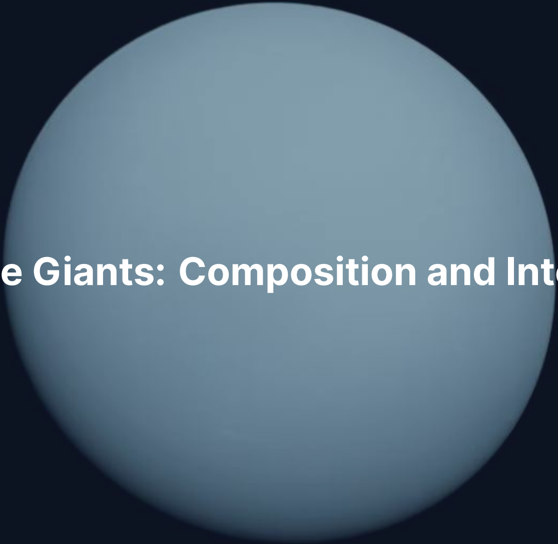
- Outer irregular moon, retrograde, highly inclined.
- Colour, density, surface chemistry → captured KBO (Agnor & Hamilton 2006).
- Source of dark dust on Iapetus's leading hemisphere.
- A free trans-Neptunian sample, accessible without leaving Saturn.

Break

End of Part 1 (Gas Giants)

Part 2: Ice Giants Uranus and Neptune

8. Ice Giants: Composition and Interior



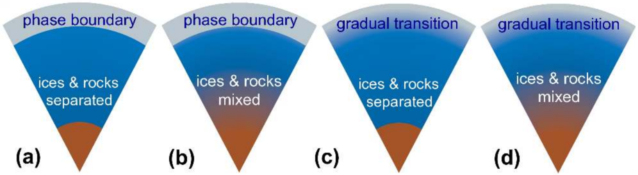
Uranus, Voyager 2 (1986). NASA/JPL-Caltech

Ice Giants: The Exotic Twins

- Uranus: 14.5 $M_\oplus$, 4.0 $R_\oplus$.
- Neptune: 17.1 $M_\oplus$, 3.9 $R_\oplus$.
- H/He envelope only 10–20% of mass.
- Bulk: water, ammonia, methane in compressed-fluid form.
- ”Ice” reflects formation as solid grains; current state is fluid.
- One spacecraft each: Voyager 2 in Jan 1986 / Aug 1989. No return mission since.

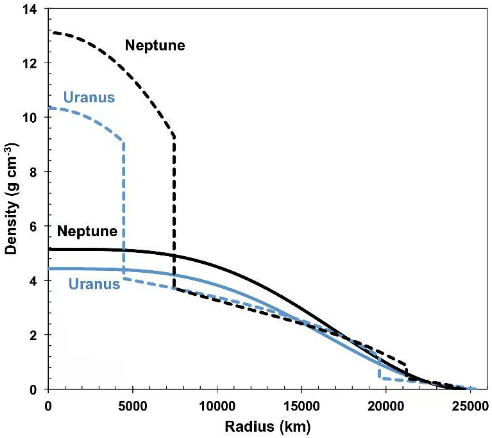
Most under-explored major planets. Knowledge ~35–40 yr out of date.

Possible Interior Structures



- (a) Sharp boundaries.
- (b) Sharp envelope, gradual ice/rock.
- (c) Gradual envelope, sharp ice/rock.
- (d) Fully gradual gradient.
- Voyager J_2, J_4 alone cannot distinguish.
- Helled+ (2020).

Density Profile Degeneracy



- Uranus (blue) and Neptune (black).
- Solid: empirical; dashed: 3-layer.
- Both fit Voyager gravity equally well.
- Cannot uniquely solve interior with current data.
- Motivates a dedicated orbiter.

Superionic Ice

- Millot+ (2019): laser shock to 100–400 GPa, several thousand K.
- In-situ X-ray diffraction → rigid bcc oxygen lattice with mobile protons.
- Material is part solid, part liquid; *ionically* conducting.
- Can sustain a planetary dynamo without metallic hydrogen.
- Soderlund+ (2020): thin-shell dynamo → multipolar, off-axis fields.

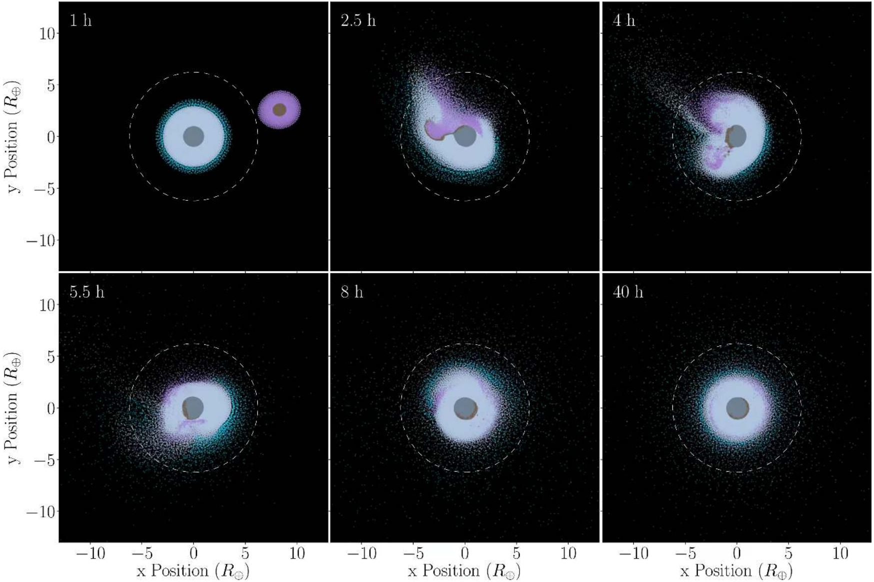
Superionic ice is the leading dynamo source for Uranus and Neptune.

Ice Giant Magnetic Fields

- Uranus: dipole tilt ~ 59° from rotation axis, offset ~ $1/3R_p$ from centre.
- Neptune: dipole tilt ~ 47°, similarly offset.
- Strong quadrupole + higher-order components dominate.
- Very different from dipole-dominated Jupiter and Saturn.
- Connerney+ 1991.
- Best explanation: thin convecting shell of ionic / superionic fluid.

9. Uranus: The Tilted Planet

Uranus clouds, Voyager 2 (PIA18182)

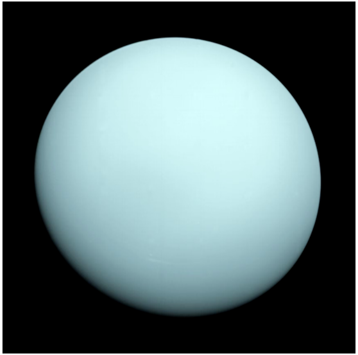


SPH simulation of giant impact on proto-Uranus. Kegerreis et al. (2018).

Reading the Uranus-Tilt Simulation

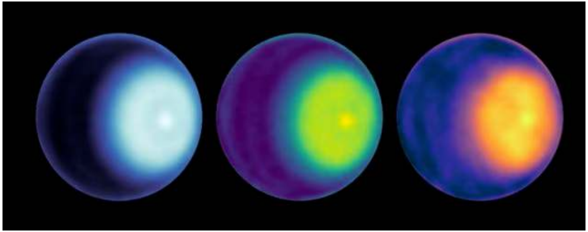
- Axial tilt 97.8°: rotation axis nearly in orbit plane.
- Each pole has 42 yr of daylight then 42 yr of night.
- Most plausible cause: oblique giant impact in late accretion.
- Snapshots (t = 1 to 40 h): light/dark grey = target ice/rock; blue = target H/He.
- Same impact must have produced equatorial debris disk that later re-accreted to satellites (Morbidelli+ 2012).

Uranus: Featureless at Voyager, Active Now



Voyager (1986) saw Uranus near solstice → pole-on, muted cloud bands.

Modern Uranus: Storms and Cyclones



- Approaching equinox: more cloud activity each year.
- 2014 storm system: amateur-detectable.
- JWST 2023 (De Pater+ 2022): vivid rings, polar cap structure.
- North-polar cyclone confirmed by VLA + JWST.

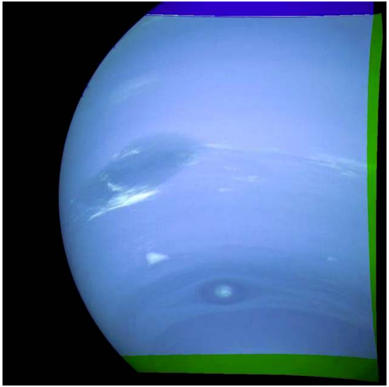
Uranus: Why So Cold Inside?

- Voyager IRIS: emits only ~ 1.06× absorbed solar flux (Pearl+ 1990).
- Neptune at similar distance: ~ 2.6×. Why is Uranus so quiet?
- Hypothesis 1: giant impact left stably stratified, inhomogeneous interior; convection inhibited, primordial heat trapped.
- Hypothesis 2: latent heat from condensation of $\mathrm{CH_4}$ or $\mathrm{H_2O}$ at depth masks deep flux.
- Hypothesis 3: external perturbation reset thermal history.
- Open question; demands dedicated orbiter.

10. Neptune and Triton

Neptune Great Dark Spot, Voyager 2 (1989). NASA/JPL-Caltech

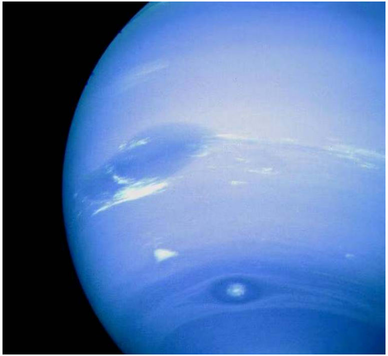
Neptune: The Active Ice Giant



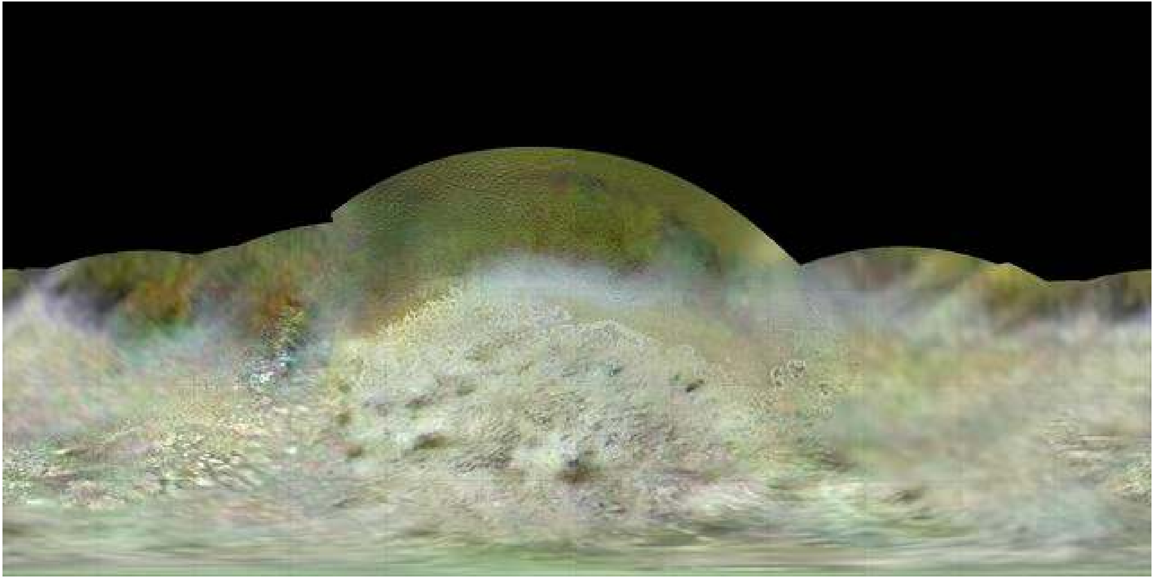
Great Dark Spot, Voyager 2 (1989)

- Active anticyclonic storm at flyby.
- GDS comparable in size to Earth.
- Bordered by bright methane cirrus.
- By 1994 (HST): GDS had vanished.
- Subsequent dark spots come and go on decade timescales.

Neptune: Fastest Winds in the Solar System



- ”Scooter”: small bright cloud at ~380 m s^{-1}.
- Equatorial easterly jet peaks ~580 m s^{-1}.
- Yet receives only 1/900 of Earth’s insolation.
- Energy must come from interior.
- Internal flux ~2.6× absorbed (Pearl & Conrath 1991).
- Source mechanism: still unresolved.



Triton’s southern hemisphere, Voyager 2 mosaic. NASA/JPL-Caltech.

Reading Triton

- Largest moon of Neptune; *retrograde* orbit ($i \approx 157^\circ$ relative to equator).
- Captured Kuiper Belt object: orbit cannot be primordial.
- Capture probably via tidal disruption of a binary near Neptune (Agnor & Hamilton 2006).
- Surface age ≲100 Myr from crater counts → recent resurfacing.
- ”Cantaloupe terrain” is unique in the solar system.
- Voyager observed 8 km tall N_2 geysers at the south pole.
- Will spiral into Roche limit on ~3.6 Gyr timescale.

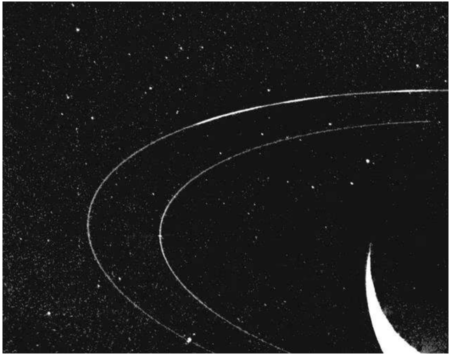
11. Ice Giant Rings

Neptune ring system, Voyager 2 (1989). NASA/JPL-Caltech

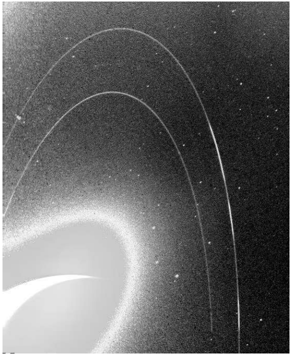
Uranus Rings: Narrow, Dark, Discovered by Occultation

- Discovered 1977 from a stellar occultation: 5+5 dimming events.
- Voyager 2 (1986): 11 rings; HST (2003–2005): 13 total.
- Most prominent: ε ring at ~51 000 km.
- All narrow ($\lesssim$10 km), dark, carbon-rich.
- Some confined by small ”shepherd” moonlets.

Neptune Rings: Arcs Held by Resonance



Adams ring + Le Verrier ring



Long-exposure forward-scattered view

Adams ring contains 5 distinct *arcs*, gravitationally trapped by mean-motion resonance with Galatea.

Why the Compositional Difference?

- Saturn's rings: >95% bright water ice.
- Uranus + Neptune rings: dark, carbon-rich.
- Different formation pathway likely: disruption of small inner moons, not large external sources.
- JWST 2023: revealed previously undetected fine structure at Uranus.
- Brighter Adams ring arcs have visibly faded since Voyager.

12. Comparative Payoff and Exploration Frontier

Europa Clipper concept. NASA/JPL-Caltech

Why Gas Giants and Ice Giants Diverged

- All four began as solid cores accreting disk gas (L2).
- For runaway gas accretion: core $\gtrsim 10\ M_\oplus$ *while gas is still present*.
- Disk lifetime 3–5 Myr is the deadline.
- Jupiter, Saturn: cores reached the threshold early enough.
- Uranus, Neptune: too late, or low-density region → stalled at 10–20 $M_\oplus$.
- Nice model (Tsiganis+ 2005): ice giants migrated and scattered planetesimals after dispersal.

Ice-giant masses are a natural intermediate outcome of core accretion in the outer disk.

Common Themes Across All Four Giants

1. Internal heat excess (Uranus is the outlier).
2. Strong zonal jets; speed *increases* with distance from Sun.
3. Global magnetic fields, but very different morphologies (dipole vs. multipole).
4. Moon and ring systems, but different inventories: bright icy moons (Saturn) vs. captured KBOs (Triton).
5. All four are laboratories for sub-Neptune and Neptune-mass exoplanets (L13).

What We Still Don’t Know

- Heavy-element fraction of Uranus and Neptune?
- What sustains Neptune’s heat flow, and why is Uranus so quiet?
- How and when did Saturn’s rings form? How long will they last?
- Origin of Jupiter’s dilute core?
- Does Callisto have a true ocean today?
- Deep structure of Saturn’s hexagonal jet?
- Habitability of Enceladus, Europa, and Titan?

Two Ocean-World Missions to Jupiter

Europa Clipper (NASA, launched Oct 2024)

- **Europa Clipper:** ~50 close flybys of Europa from Jovian orbit.
- REASON radar, magnetometer, mass spec, thermal imager.
- **JUICE** (ESA, launched Apr 2023): orbits Ganymede in 2034.
- First-ever orbiter of any moon other than Earth's.
- Complementary, not redundant.

Future: Dragonfly and a Uranus Orbiter

- Dragonfly (NASA, launch 2028, arrival 2034): first rotorcraft on another world.
- Selk crater on Titan: organics + transient melt-water from impact heat.
- Mass spec orders of magnitude better than Cassini's.
- 2022 Decadal Survey: **Uranus orbiter + probe** = top large flagship for the 2030s.
- Jupiter gravity assist → orbital insertion + atmospheric probe.
- Launch in early 2030s → arrival late 2030s / early 2040s.

The Voyager Legacy

- Launched 1977; only spacecraft ever to visit Uranus (1986) and Neptune (1989).
- Voyager 1: directed out of ecliptic after Saturn.
- Both still operational in interstellar space; expected loss of contact ~2030.
- Uranus and Neptune flyby data still yielding new science.
- Lesson 1: a single short flyby cannot substitute for sustained investigation.
- Lesson 2: well-designed missions return science long after the spacecraft is gone.

Summary: Today's Course-Level Takeaways

1. Gas giants and ice giants form a compositional continuum, set by formation timing.
2. Both gas giants now appear to host *dilute / fuzzy* cores, not compact ones.
3. Saturn's spectacular rings are young (~100 Myr) and transient.
4. Galilean moons and Saturnian moons span unprecedented diversity: Io, Europa, Ganymede, Callisto, Titan, Enceladus.
5. Ice giants are the most under-explored major planets; one flyby each, ~35 yr ago.
6. Roche limit explains where rings exist and where moons cannot accrete: $\approx 2.46\, R_p(\rho_p/\rho_s)^{1/3}$.

Questions?

Next: **Lecture 12. Small Bodies: Meteorites, Asteroids, Comets**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`